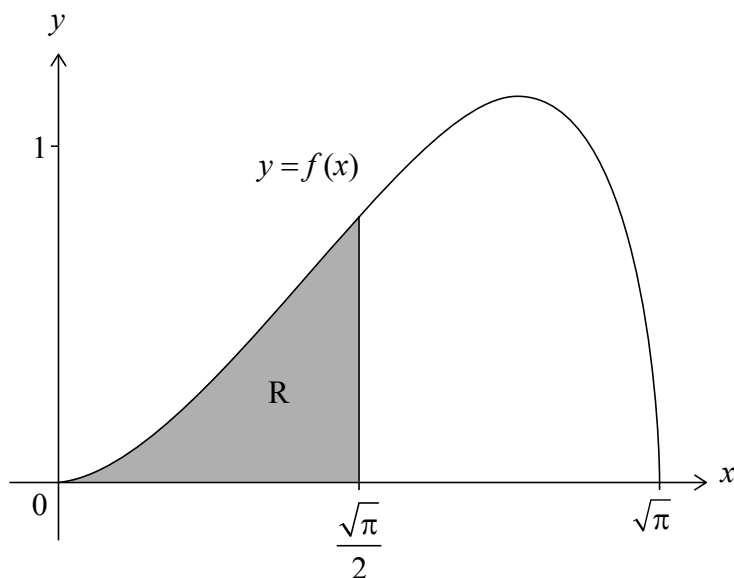


6. [Maximum mark: 6]

The function f is defined as $f(x) = \sqrt{x \sin(x^2)}$, where $0 \leq x \leq \sqrt{\pi}$.

Consider the shaded region R enclosed by the graph of f , the x -axis and the line $x = \frac{\sqrt{\pi}}{2}$, as shown in the following diagram.



The shaded region R is rotated by 2π radians about the x -axis to form a solid.

Show that the volume of the solid is $\frac{\pi(2 - \sqrt{2})}{4}$.

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